

Laboratory Report

Module V

Hands-on Labs: Building AI Agents

Building a Knowledge Retrieval Agent with LangChain · Adding Tools and Memory to Agents · Building a Workflow Agent and Multi-Agent Collaboration

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Frameworks LangChain 1.4.0 · LangGraph 1.2.11
Model gemini-2.5-flash (Google AI Studio free tier)
Language Python 3.11
Duration 9 hours (3 sessions × 3 hours)
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Chapter 1

Module Overview

1.1 Module Information

Item	Detail
Module	V – Hands-on Labs: Building AI Agents
Duration	9 hours (typically 3 sessions of 3 hours)
Prerequisite	Ability to write Python functions, loops and classes. No prior AI, ML or NLP coursework is assumed.
Cost to student	Zero. Free Google AI Studio API key; no credit card required.
Model used	<code>gemini-2.5-flash</code> (free tier) via the Gemini API
Frameworks	LangChain 1.4.0 and LangGraph 1.2.11
Hardware	Any laptop with 4 GB RAM and internet. No GPU required.
Deliverables	Working code for 8 experiments, observation notebook, record notebook, and one mini-project

1.2 Session Plan

Session	Lab	Title
1	0	Environment setup and verification
1	1	Talking to a model; why a model alone is not an agent
1	2	Knowledge retrieval pipeline (load, split, embed, store, retrieve)
2	3	Your first agent: retrieval as a tool
2	4	Adding tools: calculator, clock, Wikipedia, web search
2	5	Adding memory: short-term and long-term
3	6	Workflow agent with LangGraph: route, review, revise
3	7	Multi-agent collaboration: the supervisor pattern

1.3 Module Outcomes

Upon completion of this module the student will be able to:

- CO1** Explain the difference between a language model, a retrieval pipeline and an agent, and justify which one a given problem needs.
- CO2** Build a knowledge retrieval pipeline over a private document set and measure whether retrieval is returning the right chunks.

- CO3** Write custom tools, expose them to a model, and diagnose why a model chose or skipped a tool.
- CO4** Add short-term and long-term memory to an agent and explain the security consequence of getting thread isolation wrong.
- CO5** Design a multi-step workflow as a state graph with conditional branching and a bounded revision loop.
- CO6** Decompose a task across multiple cooperating agents under a supervisor and reason about where the pattern fails.

1.4 Important Note on API Versions

LangChain changed substantially at version 1.0. Most online tutorials still use the older API (`initialize_agent`, `AgentExecutor`, `ConversationBufferMemory`, `LLMChain`). None of those appear in this manual and none of them will work with the installed versions. All code in this report uses the current LangChain 1.x / LangGraph API.

Chapter 2

Experiment 0 — Setting Up the Environment

2.1 Aim

To install Python, create an isolated virtual environment, obtain a free Google AI Studio API key, and prove that the machine can reach a language model and an embedding model.

2.2 Duration & Outcome

Duration: 1 hour **Outcome:** CO1

2.3 Theory Summary

Before any agent code can run, four independent pieces must be correctly installed and configured:

1. **Python 3.10+** — LangChain 1.x relies on modern type syntax that older interpreters cannot parse.
2. **Virtual environment** — A private library folder belonging only to this project. Without it, installing the module packages can silently break other projects that depend on different library versions.
3. **Libraries** — `langchain`, `langgraph`, `langchain-google-genai` and supporting packages supply the agent loop, the graph engine and the model connector.
4. **API key** — A secret string that identifies the caller to Google. The model executes on Google servers; the key is the authentication token.

The single most common cause of “it worked yesterday” failures is the absence of a virtual environment.

2.4 Procedure Highlights

1. Verify Python version with `python -version` (must be ≥ 3.10).
2. Create project folder and virtual environment:

```
1 mkdir agentic-labs
2 cd agentic-labs
```

```
3 python -m venv .venv
4
```

3. Activate the environment (platform-specific commands). Prompt must show (.venv).
4. Create `requirements.txt` with pinned versions and install:

```
1 pip install -r requirements.txt
2
```

5. Obtain a free API key from <https://aistudio.google.com/apikey> (no credit card required).
6. Store the key in a file named exactly `.env` (not `.env.txt`).
7. Create the shared `config.py` that every subsequent lab imports.
8. Run the verification script `lab0_verify.py`.

2.5 Key Design Decision: `config.py`

All labs obtain the chat model and the embedding model through a single module. When a model name is retired, only one file needs editing. The API key is never hard-coded; it is read from the environment via `python-dotenv`.

2.6 Expected Output (Abridged)

```
1 =====
2 LAB 0 - Environment check
3 =====
4 [PASS] Python 3.11 (need 3.10 or newer)
5 [PASS] import langchain
6 [PASS] import langchain_core
7 [PASS] import langgraph
8 [PASS] import langchain_google_genai
9 [PASS] import dotenv
10 [PASS] GOOGLE_API_KEY loaded (ends with ...xxxx)
11 [PASS] gemini-2.5-flash answered: 'ready'
12 [PASS] gemini-embedding-001 returned a vector of 3072 numbers
13 =====
14 RESULT: ALL CHECKS PASSED - you may start Lab 1
15 =====
```

Observation / Screenshot Placeholder

```

> python lab0_verify.py

=====
LAB 0 - Environment check
=====
[PASS] Python 3.12 (need 3.10 or newer)
[PASS] import langchain
[PASS] import langchain_core
[PASS] import langgraph
[PASS] import langchain_google_genai
[PASS] import dotenv
[PASS] GOOGLE_API_KEY loaded (ends with ...eywv)
/Users/rajendran/Desktop/agent-ai-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-flash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
Direct use of automatic function calling (AFC) in Models.generate_content is not recommended. Instead, we recommend to use AFC in Chat.send_message. Similarly, direct use of AFC in Models.generate_content_stream is not recommended. Instead, we recommend to use AFC in Chat.send_message_stream.
[PASS] gemini-3.6-flash answered: 'ready'
[PASS] gemini-embedding-001 returned a vector of 3072 numbers

=====
RESULT: ALL CHECKS PASSED - you may start Lab 1
=====

```

Figure 2.1: Successful execution of `lab0_verify.py` — all environment checks passed.

2.7 Viva Questions & Answers

Viva Question & Answer

Q1. What problem does a virtual environment solve? Give a concrete example of what breaks without one.

Answer. A virtual environment isolates the set of installed packages belonging to one project from those of every other project on the same machine. Without it, installing LangChain 1.4 for this module would overwrite an older LangChain 0.x installation required by another assignment. Both projects then break because each expects a different API surface. The virtual environment keeps each project's dependency tree private.

Viva Question & Answer

Q2. Why is the API key stored in a `.env` file rather than in `config.py`?

Answer. An API key is a secret credential. Placing it in source code risks accidental commit to a public repository; automated scanners harvest leaked keys within minutes. A `.env` file is excluded from version control (via `.gitignore`) and is loaded at runtime by `python-dotenv`, keeping the secret out of the codebase while still making it available to the program.

Viva Question & Answer

Q3. Your friend copies your project folder including `.env` and runs it. Whose quota is consumed, and why is this a problem?

Answer. The friend's requests are authenticated with *your* API key, so they consume *your* free-tier quota. This is a problem because the free tier has rate limits and daily ceilings; a second user can exhaust the allowance, causing your own legitimate runs to receive 429 RESOURCE_EXHAUSTED errors.

Viva Question & Answer

Q4. What is the difference between a chat model and an embedding model? Both were tested in Lab 0.

Answer. A chat model (`gemini-2.5-flash`) accepts a sequence of messages and produces a natural-language continuation. An embedding model (`gemini-embedding-001`) maps a piece of text to a fixed-length numerical vector such that texts with similar meaning produce vectors that point in similar directions. Chat models are used for generation; embedding models are used for semantic search.

Viva Question & Answer

Q5. What does `pip install -r requirements.txt` do that `pip install langchain` does not?

Answer. The former installs a complete, version-pinned set of packages that have been verified to work together with the code in this manual. The latter installs only the latest version of the single package named `langchain`, which may pull incompatible transitive dependencies and will almost certainly diverge from the pinned versions against which every program was tested.

Chapter 3

Experiment 1 — The Model Alone Is Not an Agent

3.1 Aim

To send prompts to a language model, observe that it has no memory and no access to tools, and identify precisely which capabilities an agent must add.

3.2 Duration & Outcome

Duration: 45 minutes **Outcome:** CO1

3.3 Theory Summary

A language model performs exactly one operation: given a sequence of tokens, it predicts the most probable next tokens. Everything that appears “intelligent” in a chatbot—remembering a name, fetching a web page, performing arithmetic—is machinery built *around* the model, not inside it.

Four fundamental limitations follow:

Limitation	Why it exists	What an agent adds
No memory	Each API call is independent	A checkpointer that replays the conversation
No live data	Model was trained once, months ago	Tools that call a clock, search engine or database
No private data	College handbook never appeared in training data	Retrieval over private documents
Unreliable arithmetic	Model predicts plausible text, does not compute	A calculator tool that evaluates expressions exactly

Message types used by the modern API:

- **SystemMessage** — standing instructions from the developer (role, tone, constraints).
- **HumanMessage** — the user’s question.
- **AIMessage** — the model’s reply (and any tool-call requests).

- **ToolMessage** — the result returned by a tool the model requested.

3.4 Key Observations from Execution

- Part C demonstrates that successive `.invoke()` calls share no state: the model cannot recall a roll number stated in the previous call.
- Part D shows that the model frequently produces incorrect arithmetic for large multiplications, confirming that prediction is not computation.
- A system message changes the *form* of the answer (bullet points, word limit) but does not enlarge the model's factual knowledge.

Observation / Screenshot Placeholder

```

=====
PART C – The model has no memory of its own
=====
/Users/rajendran/Desktop/agentc-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fla
sh' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
Turn 1 -> Noted! It looks like you're a Computer Science student (likely from the 2021 batch).

How can I help you today? Whether you need assistance with coding, coursework, project ideas, or exam prep, just let me know!
/Users/rajendran/Desktop/agentc-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fla
sh' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
Turn 2 -> I do not have access to your personal information, so I don't know your roll number.

You can usually find your roll number on:
* Your student or employee ID card
* Your admit card / hall ticket (for exams)
* Your school, college, or organization's official student/employee portal
* Official emails or admission documents sent to you by your institution

OBSERVE: turn 2 has no idea. Each .invoke() is a brand new request.
Memory is something YOU add (Lab 5), not something the model has.

```

Figure 3.1: Part C – The model has no memory. The second turn cannot recall the roll number stated in the first turn.

```

=====
PART D – Three things a plain model cannot do reliably
=====
Q: What is 84937 multiplied by 2946?
/Users/rajendran/Desktop/agentc-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fla
sh' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
A: 84,937 multiplied by 2,946 is **250,224,402**.

### Step-by-step breakdown:

$$$84,937 \times 6 = 509,622$$$
$$$84,937 \times 40 = 3,397,480$$$
$$$84,937 \times 900 = 76,443,300$$$
$$$84,937 \times 2,000 = 169,874,000$$$

Summing all the partial products:
$$$509,622 + 3,397,480 + 76,443,300 + 169,874,000 = 250,224,402$$$

Q: What is today's date?
/Users/rajendran/Desktop/agentc-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fla
sh' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
A: I don't have access to real-time information, so I can't tell you today's exact date. Please check the clock or calendar on your device!

```

Figure 3.2: Part D – Capability gaps of a plain language model: incorrect arithmetic, inability to know the current date, and lack of private college data.

3.5 Viva Questions & Answers

Viva Question & Answer

Q1. What is a token, and why does the length of a conversation affect cost?

Answer. A token is the atomic unit of text that a model reads and generates; roughly three-quarters of an English word. Billing and rate limits are expressed in tokens. Because every turn of a conversation re-sends the entire history, a long dialogue multiplies the token count (and therefore the cost) of each subsequent call.

Viva Question & Answer

Q2. Explain the role of a `SystemMessage`. Can the user see it?

Answer. A `SystemMessage` supplies standing instructions—role, tone, output format, hard constraints—that condition the model’s behaviour for the entire conversation. It is injected by the application developer and is not normally visible to the end user.

Viva Question & Answer

Q3. Why does a language model produce wrong arithmetic when a four-line Python function would be correct?

Answer. The model is a statistical next-token predictor. It has seen many arithmetic examples during training and therefore emits plausible-looking numbers, but it does not possess an internal arithmetic engine. A deterministic Python evaluator, by contrast, executes the expression exactly. This is the classic motivation for giving an agent a calculator tool.

Viva Question & Answer

Q4. What does `temperature = 0` mean? Does it guarantee identical output on every run?

Answer. Temperature scales the randomness of the next-token distribution. A value of 0 forces the model to select the highest-probability token at each step (greedy decoding), maximising reproducibility. It does *not* guarantee bit-for-bit identical output across runs, because floating-point non-determinism, batching effects and provider-side updates can still introduce small variations.

Viva Question & Answer

Q5. The model said it does not know today’s date. Is that a bug or correct behaviour? Justify.

Answer. Correct behaviour. The model’s parameters were frozen at training time; it has no access to a real-time clock. Claiming a specific date would be a hallucination. The proper remedy is to equip the agent with a `current_datetime` tool.

Chapter 4

Experiment 2 — Knowledge Retrieval Pipeline (RAG)

4.1 Aim

To build a retrieval pipeline over a private document: load, split, embed, store and search; then feed the retrieved text to a model to answer a question.

4.2 Duration & Outcome

Duration: 1 hour 15 minutes **Outcome:** CO2

4.3 Theory Summary

Retrieval-Augmented Generation (RAG) solves the private-data limitation identified in Experiment 1. Instead of hoping the model memorised a document, the application locates the relevant paragraphs and supplies them as context together with the question.

The five-stage pipeline is:

LOAD → SPLIT → EMBED → STORE → RETRIEVE → GENERATE

- **Why split?** A whole document may exceed the context window; even when it fits, embedding a single paragraph yields a sharper vector than embedding an entire book.
- **Why overlap?** A hard cut can sever a sentence (and the rule it states) across two chunks. Overlap of ≈ 100 characters ensures the complete statement survives in at least one chunk.
- **What is an embedding?** A fixed-length numerical vector such that texts of similar meaning lie close together in vector space. Cosine similarity (range $[-1, 1]$) quantifies closeness; higher values indicate greater semantic similarity.
- **Vector store used:** `InMemoryVectorStore` (ships inside `langchain-core`). Production systems substitute FAISS, Chroma, Qdrant or pgvector; the pipeline remains identical.

The handbook used in this lab is deliberately fictional. Because no model has ever seen it, any correct answer must have been produced by retrieval rather than by parametric memory.

4.4 Key Results

- Typical chunk count for the sample handbook with `chunk_size=600`, `chunk_overlap=100`: approximately 9 chunks.
- Embedding dimensionality (gemini-embedding-001): 3072.
- For the question “What happens if my attendance is 70 percent?”, the Attendance section ranks first even though the query never contains the word “condonation”—demonstrating semantic rather than keyword search.

Observation / Screenshot Placeholder

```

=====
STEP 1 – Load the document
=====
Loaded college_handbook.md: 3795 characters
A Document has two parts: .page_content and .metadata
metadata = {'source': 'college_handbook.md'}

=====
STEP 2 – Split into chunks
=====
Produced 9 chunks

--- chunk 0 ---
# Sri Venkatesa Institute of Technology – Student Handbook (Sample Data)

NOTE FOR STUDENTS: this is fictional data written for the lab. It exists so that
you can prove your agent is answering from YOUR documents and not from the
model's training data. No real model has ever seen this file.

--- chunk 1 ---
## 1. Attendance

A student must maintain a minimum of 75 percent attendance in every course to be
permitted to write the end-semester examination. Students between 65 and 74
percent may apply for condonation by paying a fee of Rs. 500 per course before
the last working day of the semester. Students

=====
STEP 3 – Turn text into vectors
=====
A question becomes a vector of 3072 numbers.
First five numbers: [0.0305, -0.0129, 0.0169, -0.0315, -0.0022]

Why this matters: two texts with similar MEANING produce vectors that
point in a similar direction, even when they share no common words.

=====
STEP 4 – Store the vectors
=====
Indexed 9 chunks.
Saved the index to handbook_index.json

```

Figure 4.1: Step 2 – Document splitting: number of chunks produced and sample content of the first chunks.

```

=====
STEP 5 – Search
=====
Question: What happens if my attendance is 70 percent?
score = cosine similarity, so HIGHER is more relevant (max 1.0)

[1] score=0.7127 ## 1. Attendance A student must maintain a minimum of 75 percent attendance in every course to be permitted to write the end-semester examination. Students bet...
[2] score=0.6814 ## 3. Laboratory Courses Every laboratory course requires a minimum of 75 percent attendance and completion of all prescribed experiments. Lab records must be ...
[3] score=0.6414 ## 2. Internal Assessment Internal assessment carries 40 marks, split as follows: two internal tests of 20 marks each, of which the better score is taken (20 m...

Notice: the question never uses the word 'condonation', but the chunk that explains condonation comes back first. That is semantic search working.

```

Figure 4.2: Step 5 – Similarity search results with cosine scores for the attendance question. The Attendance section ranks highest.

```

=====
STEP 6 – Feed the retrieved chunks to the model
=====
/Users/rajendran/Desktop/agent-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-flash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
Direct use of automatic function calling (AFC) in Models.generate_content is not recommended. Instead, we recommend to use AFC in Chat.send_message. Similarly, direct use of AFC in Models.generate_content_stream is not recommended. Instead, we recommend to use AFC in Chat.send_message_stream. If your attendance is 70 percent (which falls between 65 and 74 percent), you may apply for condonation by paying a fee of Rs. 500 per course before the last working day of the semester.

That is Retrieval Augmented Generation: retrieve first, then generate.
It is still NOT an agent, because YOU decided to search. The model was never given a choice. In Lab 3 we hand that decision to the model.

```

Figure 4.3: Step 6 – Final RAG answer generated by the model using only the retrieved handbook context.

4.5 Viva Questions & Answers

Viva Question & Answer

Q1. What is chunk overlap and what failure does it prevent? Give an example.

Answer. Chunk overlap is the number of characters (or tokens) repeated from the end of one chunk at the beginning of the next. It prevents a sentence that straddles a chunk boundary from being split into two incomplete fragments, neither of which contains the full rule. Example: the sentence stating the 75% attendance threshold could be cut after “75 percent”; with overlap the complete sentence appears wholly inside at least one chunk.

Viva Question & Answer

Q2. Two sentences share no words but mean the same thing. Why does vector search find both?

Answer. Embedding models are trained so that texts of similar meaning produce vectors that point in similar directions, independent of surface lexical overlap. Cosine similarity therefore ranks the two sentences as near neighbours even when they share almost no tokens.

Viva Question & Answer

Q3. What is cosine similarity? What range does it take and which end is “more similar”?

Answer. Cosine similarity is the cosine of the angle between two vectors: $\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}$. Its range is $[-1, 1]$. Values near +1 indicate high semantic similarity; values near 0 indicate orthogonality (unrelated meanings); values near -1 indicate opposite meanings.

Viva Question & Answer

Q4. Why is the handbook in this lab fictional? What would be wrong with using a real, famous document?

Answer. A fictional handbook guarantees that the model has never encountered the text during pre-training. Consequently any correct answer must have been retrieved from the supplied document. A famous real document (e.g. the Indian Constitution) may already reside in the model’s parametric knowledge, making it impossible to prove that retrieval, rather than memorisation, produced the answer.

Viva Question & Answer

Q5. This experiment is RAG but not an agent. What exactly is missing?

Answer. The decision to search is hard-coded in the Python script. The model is never offered a choice; it merely summarises whatever context the programmer already retrieved. An agent would receive a description of the search tool and decide at run time whether (and with what query) to invoke it.

Viva Question & Answer

Q6. Where would you store the vectors if the document set were 50,000 pages instead of 4?

Answer. An in-memory store would exhaust RAM. A persistent, disk-backed or server-backed vector database such as FAISS (with on-disk index), Chroma, Qdrant or pgvector would be required. The five-stage pipeline remains unchanged; only the storage engine is replaced.

Chapter 5

Experiment 3 — Your First Agent: Retrieval as a Tool

5.1 Aim

To convert the retrieval pipeline of Experiment 2 into a tool, give it to an agent, and observe the agent deciding for itself whether and what to search.

5.2 Duration & Outcome

Duration: 1 hour **Outcomes:** CO2, CO3

5.3 Theory Summary

The decisive shift from Experiment 2 to Experiment 3 is the transfer of the search decision from the programmer to the model. The model is informed of the existence of a tool and, at run time, chooses whether to call it.

The agent loop implemented by `create_agent` consists of four repeated steps:

1. Send the conversation plus tool descriptions to the model.
2. If the model replies with ordinary text, that is the final answer; stop.
3. If the model replies with a tool call, execute the tool.
4. Append the tool result to the conversation and return to step 1.

When a function is decorated with `@tool`, three pieces of information are automatically exposed to the model:

1. the function name,
2. the type hints (which become the argument schema),
3. the docstring (the sole instruction telling the model *when* to use the tool).

A weak docstring such as “Searches the handbook” produces unreliable tool selection. A strong docstring enumerates the topics covered and explicitly states the situations in which the tool must be invoked.

5.4 Key Observations

- For the attendance question the model rewrote the user’s wording into a richer query containing the term “condonation”—a term never supplied by the user.
- For the greeting “Hello, who are you?” the model issued zero tool calls, correctly judging that search was unnecessary.
- The system prompt instruction “Always call `search_handbook` before answering ...” is essential; without it the model frequently answers from parametric knowledge.

Observation / Screenshot Placeholder

```
=====
Q: I have 70 percent attendance. Can I write the exam?
/Users/rajendran/Desktop/agent-ic-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fl
ash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
/Users/rajendran/Desktop/agent-ic-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fl
ash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
[tool returned] call search_handbook({'query': 'attendance requirements for writing exam'})
[tool returned] [section from college_handbook.md] ## 1. Attendance A student must maintain a minimum of 75 percent attendance in every...

ANSWER: You cannot directly write the end-semester exam, as a minimum of 75 percent attendance is required. However, since your attendance is between
65 and 74 percent, you may apply for condonation by paying a fee of Rs. 500 per course before the last working day of the semester.

=====
Q: How late can I keep a library book before I start paying?
/Users/rajendran/Desktop/agent-ic-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fl
ash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
/Users/rajendran/Desktop/agent-ic-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fl
ash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
[tool returned] call search_handbook({'query': 'library book return late fee fine days limit overdue'})
[tool returned] [section from college_handbook.md] ## 4. Library Each student may borrow a maximum of four books at a time for a period...

ANSWER: You can keep a borrowed book for up to 14 days. It can be renewed once for an additional 14 days if no other student has reserved it. After th
is period, an overdue fine of Rs. 2 per day per book will be charged.

=====
Q: What is the Wi-Fi password in the hostel?
/Users/rajendran/Desktop/agent-ic-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fl
ash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
/Users/rajendran/Desktop/agent-ic-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fl
ash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
[tool returned] call search_handbook({'query': 'hostel Wi-Fi password'})
[tool returned] [section from college_handbook.md] ## 5. Hostel Hostel gates close at 9:30 pm on weekdays and 10:00 pm on weekends. Lat...

ANSWER: The student handbook does not state the hostel Wi-Fi password; it only mentions that Wi-Fi is available in common areas and personal routers a
re prohibited. Please ask your class advisor for assistance.
```

Figure 5.1: Agent execution traces for multiple questions. Notice the model deciding when to call `search_handbook`, the rewritten search queries, and the final grounded answers.

5.5 Viva Questions & Answers

Viva Question & Answer

Q1. Name the three things the `@tool` decorator sends to the model.

Answer. (1) the function name, (2) the type hints of its parameters (which become the JSON schema the model must fill), and (3) the docstring, which is the only natural-language instruction the model receives about when the tool should be used.

Viva Question & Answer

Q2. What exactly makes this program an agent, when Experiment 2 was not one?

Answer. In Experiment 2 the Python code always performed retrieval; the model never chose. In Experiment 3 the model receives a tool description and autonomously decides, on each turn, whether to call the tool, what query to supply, and when it has obtained enough information to answer. That run-time decision constitutes agency.

Viva Question & Answer

Q3. The model searched for “condonation” although the student never used that word. How?

Answer. The model, having been told the tool covers attendance rules, reformulated the user’s informal question into a more precise retrieval query that maximises the chance of hitting the relevant handbook section. This query rewriting is an emergent behaviour of the agent loop, not hard-coded logic.

Viva Question & Answer

Q4. Why does the system prompt instruct the agent to say when something is not covered? What would it do otherwise?

Answer. Without an explicit grounding instruction the model tends to fill gaps with plausible-sounding general knowledge (hallucination). The instruction forces the model to admit the limits of the retrieved material and to direct the student to a human advisor, preserving trustworthiness.

Viva Question & Answer

Q5. Describe the four steps of the agent loop.

Answer. (1) Send conversation + tool schemas to the model; (2) if the reply is plain text, terminate with that text as the answer; (3) if the reply contains a tool call, execute the named tool with the supplied arguments; (4) append the tool result as a `ToolMessage` and repeat from step 1.

Viva Question & Answer

Q6. The agent made no tool call for “Hello, who are you?”. Is that correct behaviour? Justify.

Answer. Yes. The query is a pure identity/greeting request that does not concern college rules. Calling the handbook tool would waste tokens and latency. Correct agency includes the judgement that a tool is *not* required.

Chapter 6

Experiment 4 — Adding Tools to an Agent

6.1 Aim

To build four tools of different kinds, expose them all to a single agent, and observe the agent selecting between them and chaining them.

6.2 Duration & Outcome

Duration: 1 hour 15 minutes **Outcome:** CO3

6.3 Theory Summary

A tool is any Python function the model is permitted to call. The four tools deliberately address four distinct capability gaps:

Tool	Kind	Limitation removed
<code>calculate</code>	Pure computation	Model predicts arithmetic instead of computing it
<code>current_datetime</code>	Machine state	Model has no notion of “today”
<code>search_wikipedia</code>	External library	Model knowledge is frozen at training cut-off
<code>search_web</code>	Live network	Model cannot see information published after training

Security note. The calculator must never use `eval()`. Model-generated strings can contain arbitrary Python (including `os.system` calls). The implementation therefore parses the expression into an AST and walks only a whitelist of arithmetic operators.

Tool-design checklist.

- One clear responsibility per tool.
- Docstring states *when* to use the tool, not merely what it does.
- Type hints on every parameter.
- Return a readable string; never raise an unhandled exception.

- Treat every argument as untrusted input.

6.4 Key Observations

- The multiplication that previously failed is now answered correctly by the calculator tool.
- The final question (“latest Python version times 100”) required a two-step chain: web search followed by calculation on the retrieved number. The sequence was discovered by the agent loop, not hard-coded.

Observation / Screenshot Placeholder

```

> python lab4_tools_agent.py
Tools registered:
  calculate      Evaluate an arithmetic expression and return the exact result. Use thi
  current_datetime Return today's date and the current time. Use this whenever the questi
  search_wikipedia Look up a factual topic on Wikipedia and return a short summary. Good
  search_web     Search the live web and return the top results with their URLs. Use th

=====
Q: What is 84937 multiplied by 2946?
=====
/Users/rajendran/Desktop/agent-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fl
ash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
/Users/rajendran/Desktop/agent-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fl
ash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
-> calling calculate({'expression': '84937 * 2946'})
<- 250224402

ANSWER: According to the 'calculate' tool, 84937 multiplied by 2946 is **250,224,402**.

=====
Q: Who invented the transistor, and in what year?
=====
/Users/rajendran/Desktop/agent-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fl
ash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
/Users/rajendran/Desktop/agent-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fl
ash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
/Users/rajendran/Desktop/agent-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fl
ash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
/Users/rajendran/Desktop/agent-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fl
ash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
/Users/rajendran/Desktop/agent-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fl
ash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
-> calling search_wikipedia({'topic': 'Transistor'})
<- Wikipedia lookup failed: Expecting value: line 1 column 1 (char 0)
-> calling search_web({'query': 'who invented the transistor in what year'})
<- History of the transistor - Wikipedia 5 days ago - This Month in Physics History: November 17 to December 23,
-> calling search_web({'query': 'John Bardeen Walter Brattain William Shockley point contact transistor 1947 Bell Labs'})
<- Point-contact transistor - Wikipedia The point-contact transistor was the first type of transistor to be succe
-> calling search_web({'query': 'Julius Edgar Lilienfeld transistor patent year'})
<- Julius Edgar Lilienfeld - Wikipedia Julius Edgar Lilienfeld (April 18, 1882 - August 28, 1963) was an Austro-H

ANSWER: The first working transistor was invented in **December 1947** at AT&T's Bell Laboratories by **John Bardeen** and **Walter Brattain**, under
the direction of solid-state physics group leader **William Shockley**.

Here is the breakdown of the major milestones:

### 1. The Point-Contact Transistor (1947)
* **Inventors:** John Bardeen and Walter Brattain
* **Year:** 1947 (first successfully demonstrated on December 16, 1947, and presented internally at Bell Labs on December 23, 1947)

```

Figure 6.1: Tool selection traces across multiple questions. Observe the correct choice of tools and the multi-step chain (web search followed by calculation) in the final question.

6.5 Viva Questions & Answers

Viva Question & Answer

Q1. Why must a tool never use `eval()` on model-supplied input?

Answer. The expression string is generated by a language model that can be influenced by adversarial text appearing in retrieved documents or user prompts. `eval()` would execute arbitrary Python, including destructive system commands. A safe calculator therefore restricts evaluation to a closed set of arithmetic operators via AST inspection.

Viva Question & Answer

Q2. What does the type hint on a tool parameter accomplish?

Answer. The type hint is converted into a JSON schema that is presented to the model. The model is thereby constrained to supply arguments of the expected type and structure, reducing malformed tool calls.

Viva Question & Answer

Q3. A tool raises an unhandled exception. What happens to the agent run, and how should you have written the tool?

Answer. An unhandled exception aborts the entire agent loop. Every tool body should be wrapped in a broad `try/except` that returns a descriptive error string; the model can then decide how to proceed.

Viva Question & Answer

Q4. How did the agent know to call `search_web` before `calculate` in the last question?

Answer. The system prompt instructed the model never to perform arithmetic mentally and to prefer web search for recent facts. After receiving the version number from the search tool, the model recognised that a calculation was still required and issued a second tool call. The ordering emerged from successive decisions inside the agent loop.

Viva Question & Answer

Q5. You have twenty tools and the agent starts choosing badly. Give two ways to fix this.

Answer. (1) Strengthen and specialise each docstring so that the conditions of use are unambiguous; (2) partition the tools among several specialist agents coordinated by a supervisor (the pattern of Experiment 7), thereby giving each agent a short prompt and a small tool set.

Viva Question & Answer

Q6. What is the difference between a tool and a function? Be precise.

Answer. A function is ordinary Python callable code. A tool is a function that has been registered with the agent framework (via the `@tool` decorator or equivalent) so that its name, schema and docstring become part of the model's prompt and the framework automatically executes the function when the model emits a corresponding tool-call request.

Chapter 7

Experiment 5 — Adding Memory to an Agent

7.1 Aim

To add short-term memory with a checkpointer, add long-term memory with tools, demonstrate thread isolation, and control the growth of a long conversation.

7.2 Duration & Outcome

Duration: 1 hour 15 minutes **Outcome:** CO4

7.3 Theory Summary

“Memory” in agent systems comprises two unrelated mechanisms:

	Short-term memory	Long-term memory
What it holds	Current conversation	Facts that outlive the conversation
Mechanism	Checkpointer keyed by <code>thread_id</code>	Tools that read/write persistent storage
Who decides	Automatic once configured	Agent decides what is worth saving
Lifetime	Process lifetime (InMemorySaver)	Until explicitly deleted
Analogy	What you remember during a phone call	What you write in a notebook afterwards

Security implication of `thread_id`. The same `thread_id` recovers the same conversation; a different `thread_id` starts a clean slate. In a multi-user application the `thread_id` must be derived from a user identity (roll number, session id). Sharing a `thread_id` across users leaks private conversation history—one of the most common defects in student and production chatbots.

Conversation growth. Every turn re-sends the entire history. `SummarizationMiddleware` monitors message count and, past a threshold, replaces older messages with a concise summary while retaining the most recent turns verbatim.

7.4 Key Observations

- Thread “student-A” retains the student’s name and semester; thread “student-B” (same agent object) does not.
- After a process restart, long-term facts stored in `long_term_memory.json` are still available on the first turn via the `recall_facts` tool.
- With summarisation enabled, the retained message count stabilises even after many turns.

Observation / Screenshot Placeholder

```

=====
PART A - Short-term memory
=====
--- thread 'student-A' ---
You : My name is Divya and I am in the 5th semester, ECE branch.
/Users/rajendran/Desktop/agent-ic-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fl
ash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
Direct use of automatic function calling (AFC) in Models.generate_content is not recommended. Instead, we recommend to use AFC in Chat.send_message. S
imilarly, direct use of AFC in Models.generate_content_stream is not recommended. Instead, we recommend to use AFC in Chat.send_message_stream.
Bot : Welcome, Divya! It's great to connect with a 5th-semester ECE student. This is a crucial year for core subjects and project planning. How can I
help you with your academics, electives, or career goals today?

You : Which semester am I in?
/Users/rajendran/Desktop/agent-ic-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fl
ash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
Bot : You are currently in your 5th semester of the ECE program! Is there anything specific about this semester's coursework or schedule you'd like to
discuss?

--- thread 'student-B' (a different student, same agent object) ---
You : Which semester am I in?
/Users/rajendran/Desktop/agent-ic-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fl
ash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
Bot : Hi there! I don't have access to your personal student records. You can check your student portal, or tell me when you started and your complete
d credits, and we can figure it out together!

OBSERVE: thread 'student-A' remembers. Thread 'student-B' does not,
because it is a separate conversation. In a real application the
thread_id would be the student's roll number or a chat session id.
Getting this wrong is how chatbots leak one user's data to another.

Messages stored under 'student-A': 4

```

Figure 7.1: Part A – Short-term memory with checkpoint. Thread `student-A` retains context while thread `student-B` remains isolated.

```

=====
PART B - Long-term memory
=====
You : Hi, I am doing my final year project on solar panel fault detection.
/Users/rajendran/Desktop/agent-ic-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fl
ash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
/Users/rajendran/Desktop/agent-ic-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fl
ash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
[tool] recall_facts({})
[tool] remember_fact({'value': 'solar panel fault detection', 'key': 'project_topic'})
Bot : That sounds like a great final year project! Solar panel fault detection has excellent practical applications. How can I help you get started or
structure your work?

You : What did I say my project was about?
/Users/rajendran/Desktop/agent-ic-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fl
ash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
[tool] recall_facts({})
[tool] remember_fact({'value': 'solar panel fault detection', 'key': 'project_topic'})
Bot : You mentioned that your final year project is on solar panel fault detection. How is it coming along?

Contents of long_term_memory.json right now:
{
  "project_topic": "solar panel fault detection"
}

Run this script again. Part B will recall the project topic on the
FIRST turn, because it came from the file, not from the conversation.

```

Figure 7.2: Part B – Long-term memory. Facts are persisted to `long_term_memory.json` and can be recalled across process restarts.

7.5 Viva Questions & Answers

Viva Question & Answer

Q1. Distinguish short-term and long-term memory in an agent. Give a mechanism for each.

Answer. Short-term memory holds the ongoing conversation and is implemented by a checkpointer (e.g. `InMemorySaver`) keyed by `thread_id`. Long-term memory holds durable facts that must survive process restarts and is implemented by explicit tools that read from and write to a persistent store (file or database).

Viva Question & Answer

Q2. What is a `thread_id`? What goes wrong if two users share one?

Answer. A `thread_id` is the key under which a checkpointer stores conversation state. If two users share the same key, each user's messages are written into the same state object; consequently each user can see (and continue) the other's private conversation—a serious privacy breach.

Viva Question & Answer

Q3. Why does `InMemorySaver` lose everything when the program exits? When is that acceptable?

Answer. `InMemorySaver` stores state exclusively in RAM. Process termination discards the memory. It is acceptable for interactive laboratory sessions, single-user demos, and any scenario in which conversation persistence across restarts is not required.

Viva Question & Answer

Q4. Why is long-term memory implemented as tools rather than happening automatically?

Answer. Not every utterance is worth permanent storage. Automatic persistence would accumulate noise and potentially sensitive data. By exposing `remember_fact` and `recall_facts` as tools, the model itself decides, under the guidance of the system prompt, which statements constitute durable facts.

Viva Question & Answer

Q5. What problem does `SummarizationMiddleware` solve, and what does it cost you?

Answer. It prevents unbounded growth of the context window (and the associated token cost and eventual context-length errors) by periodically replacing older messages with a summary. The cost is a loss of fine-grained detail: once summarised, the exact wording of early turns is no longer available to the model.

Viva Question & Answer

Q6. You must keep conversations across a server restart. What would you change?

Answer. Replace `InMemorySaver` with a durable checkpointer backed by a database (e.g. Postgres, Redis, or a file-based saver). The rest of the agent code remains unchanged; only the checkpointer implementation is swapped.

Chapter 8

Experiment 6 — Workflow Agent with LangGraph

8.1 Aim

To build a multi-step workflow as a state graph with conditional routing, a self-review step, and a bounded revision loop.

8.2 Duration & Outcome

Duration: 1 hour 15 minutes **Outcome:** CO5

8.3 Theory Summary

`create_agent` builds a single reactive loop. When the sequence of steps is known in advance, an explicit graph is clearer and more controllable. LangGraph supplies the primitives:

- **State** — a `TypedDict` that every node reads and updates.
- **Node** — a pure Python function receiving state and returning a partial update.
- **Edge** — unconditional successor.
- **Conditional edge** — a routing function that selects the next node at run time.
- **Reducer** — the merge policy for a state key (default = replace; `add_messages` = append).

The graph constructed in this experiment implements the classic classify–route–draft–review–revise pattern. The review node performs *reflection*: a second model call judges the draft produced by the first. Reflection is one of the cheapest quality improvements available because judging finished text is easier than producing it.

Critical safety rule. Every loop must possess an explicit bound (`MAX_REVISIONS`). Without it a never-satisfied reviewer can consume an entire class API quota in minutes.

8.4 Key Observations

- Classification correctly routes academic, hostel and technical tickets to distinct desks.

- The review step frequently issues a REVISE on the first draft and accepts the improved second draft.
- The escape hatch activates when the revision counter reaches the configured maximum.

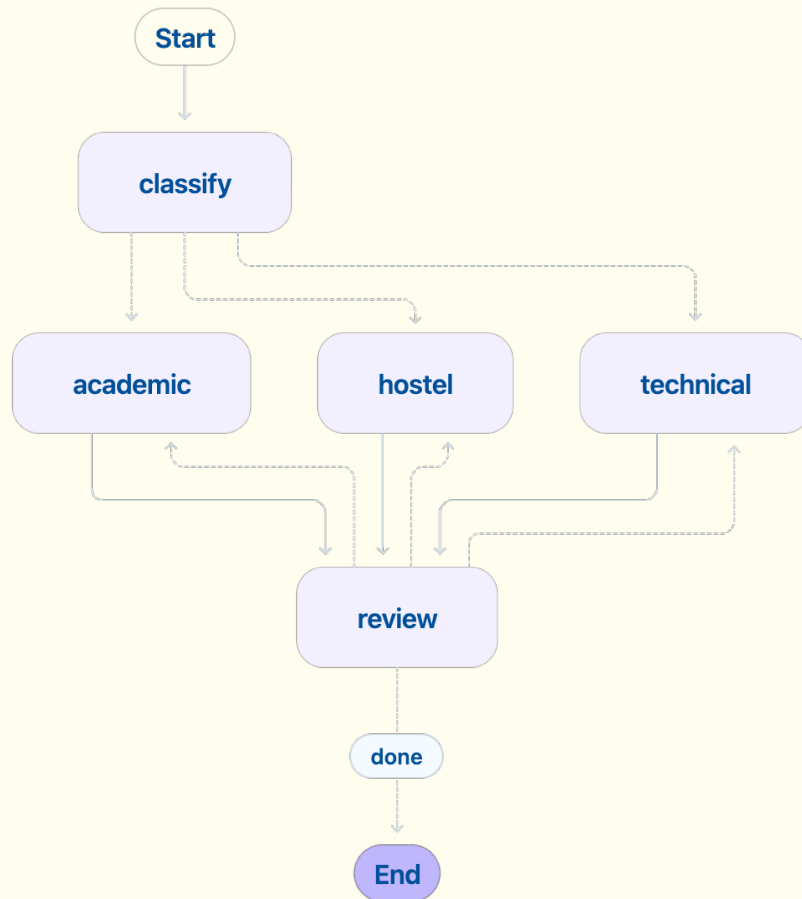
Observation / Screenshot Placeholder

Figure 8.1: Compiled LangGraph workflow. The graph shows classification, conditional routing to specialised desks, review, and a bounded revision loop.

Observation / Screenshot Placeholder

```

=====
TICKET: I was absent for two lab sessions because of fever. What do I do now?
=====
/Users/rajendran/Desktop/agent-ic-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fl
ash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
Direct use of automatic function calling (AFC) in Models.generate_content is not recommended. Instead, we recommend to use AFC in Chat.send_message. S
imilarly, direct use of AFC in Models.generate_content_stream is not recommended. Instead, we recommend to use AFC in Chat.send_message_stream.
[classify] => academic
[route] academic desk
/Users/rajendran/Desktop/agent-ic-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fl
ash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
/Users/rajendran/Desktop/agent-ic-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fl
ash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
[review] PASS

DESK      : academic
VERDICT   : PASS (after 0 revision/s)
REPLY     : Submit a written leave application along with a valid medical certificate to the Academic Office for approval.

Once approved, show the endorsed copy to your respective lab instructor immediately to request makeup lab sessions. Ensure you complete and get the mi
ssed experiments signed off before the upcoming internal evaluations. Keep a photocopy of the approved application for your records.

=====
TICKET: The mess food timing clashes with my evening class. Can I get a packed dinner?
=====
/Users/rajendran/Desktop/agent-ic-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fl
ash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
[classify] => hostel
[route] hostel desk
/Users/rajendran/Desktop/agent-ic-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fl
ash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
/Users/rajendran/Desktop/agent-ic-labs/.venv/lib/python3.12/site-packages/langchain_google_genai/chat_models.py:3908: UserWarning: Model 'gemini-3.6-fl
ash' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
[review] PASS

DESK      : hostel
VERDICT   : PASS (after 0 revision/s)
REPLY     : Yes, you can get a packed dinner. Please submit a written request along with a copy of your class timetable to the warden's office by 4:00 P
M.

Once approved, show the signed permission slip to the mess supervisor to collect your packed dinner box before leaving for your class.
=====

```

Figure 8.2: Execution traces for the three sample tickets. Each ticket follows the path classify → appropriate desk → review (with possible revision).

8.5 Viva Questions & Answers

Viva Question & Answer

Q1. Define state, node and edge in LangGraph.

Answer. *State* is the shared dictionary (normally a `TypedDict`) that flows through the graph. A *node* is a function that receives the current state and returns a partial dictionary of updates. An *edge* is a directed connection that determines which node executes next.

Viva Question & Answer

Q2. What is the difference between an edge and a conditional edge?

Answer. An ordinary edge always transfers control to a fixed successor. A conditional edge invokes a routing function that inspects the current state and returns the name of the next node, enabling branching and loops.

Viva Question & Answer

Q3. What is a reducer and when do you need one?

Answer. A reducer is the function that merges a node's return value into the existing state for a given key. The default is replacement. When a key must accumulate values (most commonly the message list), an explicit reducer such as `add_messages` is required; otherwise each node would erase the previous node's contribution.

Viva Question & Answer

Q4. Why does the review node call the model a second time instead of reusing the first reply?

Answer. Generating text and critiquing finished text are cognitively different tasks. A dedicated review call sees the complete draft as a whole and can apply explicit quality criteria, producing higher-quality final output than a single generative pass.

Viva Question & Answer

Q5. What happens if a loop has no exit condition? How does LangGraph protect you, and why should you not rely on that?

Answer. The graph would recurse indefinitely, exhausting API quota and eventually raising a recursion-limit error. LangGraph imposes a configurable recursion limit as a last-resort safety net, but the limit is a crash, not a graceful termination. Correct design places an explicit bound (counter + conditional edge to END) inside the application logic.

Viva Question & Answer

Q6. When would you choose `create_agent` over a hand-built graph, and when the reverse?

Answer. Use `create_agent` when the sequence of tool calls cannot be known in advance and the model must decide dynamically. Use a hand-built LangGraph workflow when the steps, branches and termination conditions are known a priori and must be enforced deterministically (classification pipelines, approval loops, multi-stage document processing).

Viva Question & Answer

Q7. Why is the output of `classify` normalised in Python rather than used directly?

Answer. Language models frequently emit near-miss strings (“Academic.”, “this is an academic query”). Using the raw string as a dictionary key or edge label would raise a `KeyError` or take an unintended branch. Normalising against a closed set of allowed labels and supplying a safe default makes the control flow robust.

Chapter 9

Experiment 7 — Multi-Agent Collaboration

9.1 Aim

To decompose a task across three specialist agents coordinated by a supervisor, and to observe where the pattern helps and where it fails.

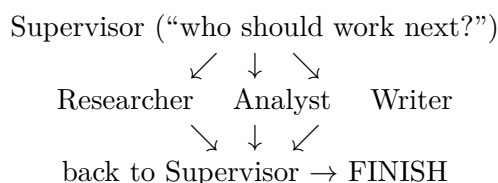
9.2 Duration & Outcome

Duration: 1 hour 15 minutes **Outcome:** CO6

9.3 Theory Summary

A single agent equipped with many tools and a long system prompt begins to choose tools poorly and to forget early instructions. Decomposing the work into specialist agents, each given a short prompt and a small tool set, measurably improves reliability on narrow sub-tasks.

The supervisor pattern:



- Workers never call one another; each reports back to the supervisor. Routing decisions therefore reside in a single, debuggable location.
- The `add_messages` reducer ensures that each worker’s report is *appended* to the shared message list rather than overwriting previous reports.
- `MAX_STEPS` bounds the number of supervisor decisions, preventing infinite ping-pong between workers.

Honest limitations. Multi-agent systems incur higher cost (extra supervisor calls), higher latency (sequential execution) and the risk of compounding error (a wrong fact from the researcher is treated as truth by later workers). The pattern is justified only when the sub-tasks genuinely require distinct tools and instructions.

9.4 Key Observations

- The supervisor typically orders the workers researcher → analyst → writer → FINISH.
- The exact order can vary slightly across runs (non-determinism is inherent).
- Arithmetic performed by the analyst can be verified by hand against the researcher's figures.

Observation / Screenshot Placeholder

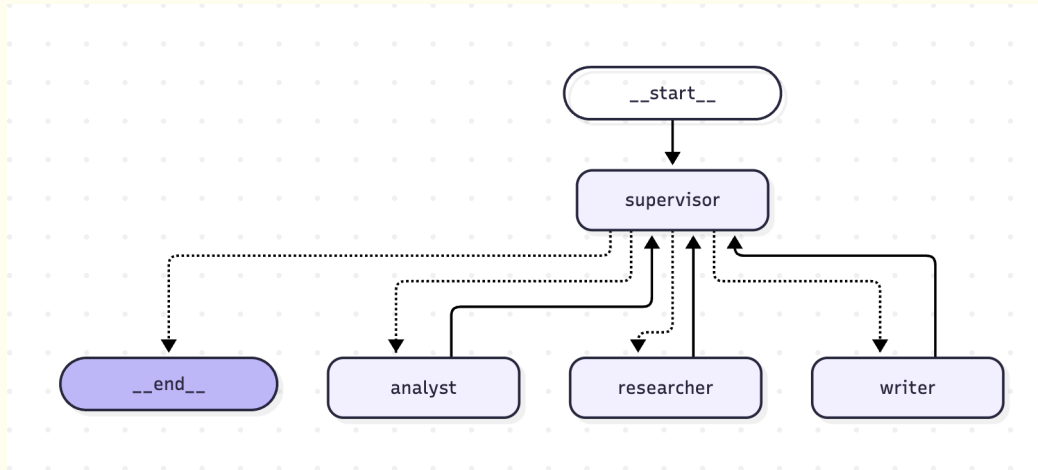


Figure 9.1: Compiled multi-agent team graph showing the supervisor and the three specialist workers (researcher, analyst, writer).

9.5 Viva Questions & Answers

Viva Question & Answer

Q1. What is the supervisor pattern? Draw it.

Answer. A central supervisor agent inspects the shared conversation and emits the name of the next specialist worker (or FINISH). Each worker performs its narrow task, appends its report to the shared state, and returns control to the supervisor. The topology is a star: all edges run between the supervisor and the workers; workers never communicate directly.

Viva Question & Answer

Q2. Why do workers report back to the supervisor instead of calling each other?

Answer. Centralising the routing decision yields a single point of control and a single point of debugging. Direct worker-to-worker calls would distribute control flow, making it harder to enforce ordering constraints, to insert global bounds, and to diagnose misbehaviour.

Viva Question & Answer

Q3. What does the `add_messages` reducer do, and what breaks without it?

Answer. It changes the merge policy for the `messages` key from “replace” to “append” (with de-duplication by message id). Without it each worker would overwrite the previous worker's report; the supervisor would see only the most recent message and lose the cumulative context required for correct routing.

Viva Question & Answer

Q4. Give two concrete costs of multi-agent versus single-agent.

Answer. (1) *Token / monetary cost*: every hand-off requires an additional supervisor model call; a four-step task may consume roughly three times as many calls. (2) *Latency*: the calls are sequential, so wall-clock time increases proportionally.

Viva Question & Answer

Q5. The researcher reports a wrong figure. What happens downstream, and how would you defend against it?

Answer. Downstream agents treat the figure as authoritative; the analyst computes on it and the writer embeds it in the final document, propagating the error. Defences include: (a) adding an explicit reviewer agent that cross-checks numbers against sources, (b) instructing the researcher to attach source URLs, and (c) inserting a verification tool that re-fetches critical statistics.

Viva Question & Answer

Q6. What does `MAX_STEPS` protect against? Name another guard you could add.

Answer. It protects against supervisor drift—an infinite or excessively long oscillation between workers. An additional guard could be a per-worker call counter that forbids the same worker from being invoked more than N times, or a wall-clock timeout on the entire graph execution.

Viva Question & Answer

Q7. Give one task where multi-agent is clearly worth it and one where it clearly is not. Justify both.

Answer. *Worth it*: producing a technical brief that requires fresh web research, quantitative analysis of the retrieved numbers, and polished writing—each sub-task benefits from a specialised prompt and tool set. *Not worth it*: answering a single factual question that can be resolved by one web-search tool call; the extra supervisor and worker overhead adds cost and latency with no quality gain.

Chapter 10

Conclusion

This laboratory module systematically constructs the conceptual and practical stack required to move from a bare language model to production-grade agentic systems:

1. A model alone is a next-token predictor; it possesses neither memory, nor tools, nor private knowledge.
2. Retrieval (RAG) injects private documents at inference time.
3. Tools convert the model into an agent that can decide, act and observe.
4. Memory (short-term via checkpointers, long-term via tools) gives the agent continuity across turns and across process restarts, while thread isolation protects user privacy.
5. Explicit state graphs (LangGraph) encode known multi-step workflows with conditional branching and bounded loops.
6. Multi-agent supervision decomposes complex tasks across specialists, trading increased cost and latency for improved reliability on heterogeneous sub-tasks.

The decisive engineering lessons are:

- Treat every model-generated tool argument as untrusted input.
- Bound every loop before writing the loop body.
- Derive `thread_id` from a stable user identity.
- Prefer a short, specialised prompt and a small tool set over a monolithic agent.
- Always inspect the full message transcript before changing code; the transcript reveals whether the failure lies in tool selection, argument formation, tool execution or final generation.

The mini-project that follows this module is the natural next step: students must assemble at least four of the six demonstrated capabilities into a system that solves a genuine campus problem, and must honestly document both successes and failure modes.

References & Resources

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4. Module V Laboratory Manual (Internal), September 2026, verified against LangChain 1.4.0 and LangGraph 1.2.11.